IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

ANALYSIS AND DESIGN OF CIRCUITS

Thursday, 30 April 2026, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Question One carries 40% of the marks. Question 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Z. Akhtar, P. Mitcheson
Second Marker(s) : A. Holmes

© Imperial College London

Information for Candidates

Numerical Answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$, while the phasor voltage is denoted by X.
- Component and source values are generally given in Ohms, Farads, Henrys, Volts, or Amps, unless otherwise stated.
- Times are in seconds, unless otherwise stated.

1. (a) For the circuit shown in Figure 1a, use the principle of superposition to determine the Thevenin equivalent circuit as seen from nodes X and Y. [6]

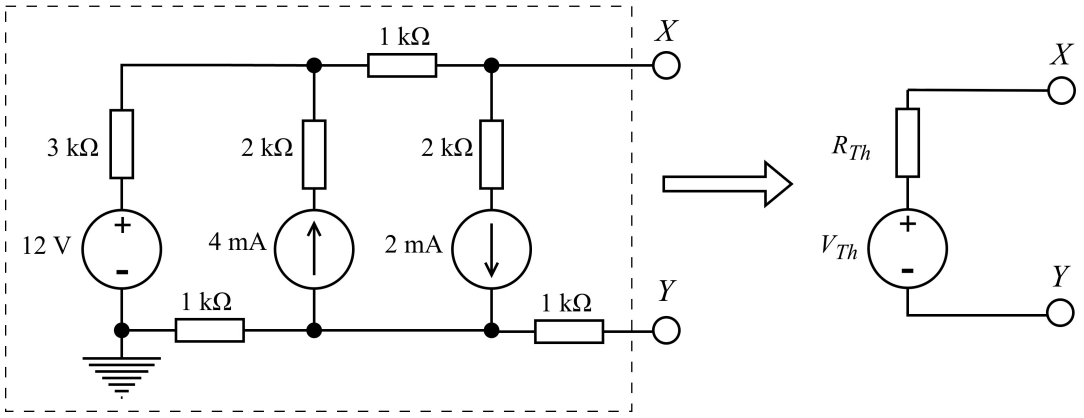


Figure 1a

(b) For the circuit shown in Figure 1b, the input voltage is [4+3]

$$V_{in}(t) = 400\cos(100\pi t) \text{ V}.$$

i. With the capacitor branch open-circuited (C_P disconnected), determine the output voltage $V_{out}(t)$, the load active power P_L, and the load reactive power Q_L.

ii. The capacitor C_P is then connected in parallel with the load. Determine the new output voltage $V_{out}(t)$. Only the steady-state voltage is required. The transient can be ignored.

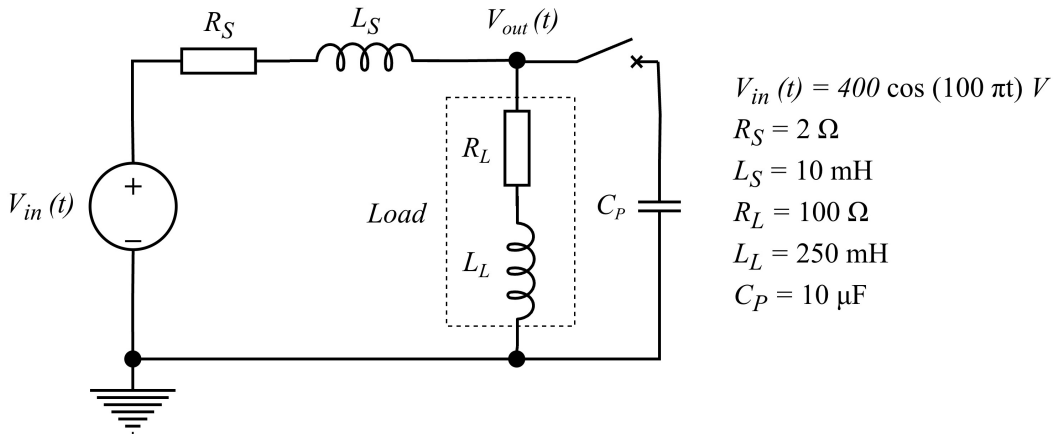


Figure 1b

(c) Figure 1c shows the circuit and the input signal used in this question. [2+5]

i. For the circuit shown in Figure 1c - (i), determine the transfer function:

$$H(j\omega) = \frac{Y}{X}$$

ii. The input voltage $x(t)$ changes from $-5\,\text{V}$ to $+10\,\text{V}$ at $t = 0$, as illustrated in Figure 1c - (ii). Assuming the circuit is in steady state for $t < 0$, determine the output voltage $y(t)$ for $t \geq 0$.

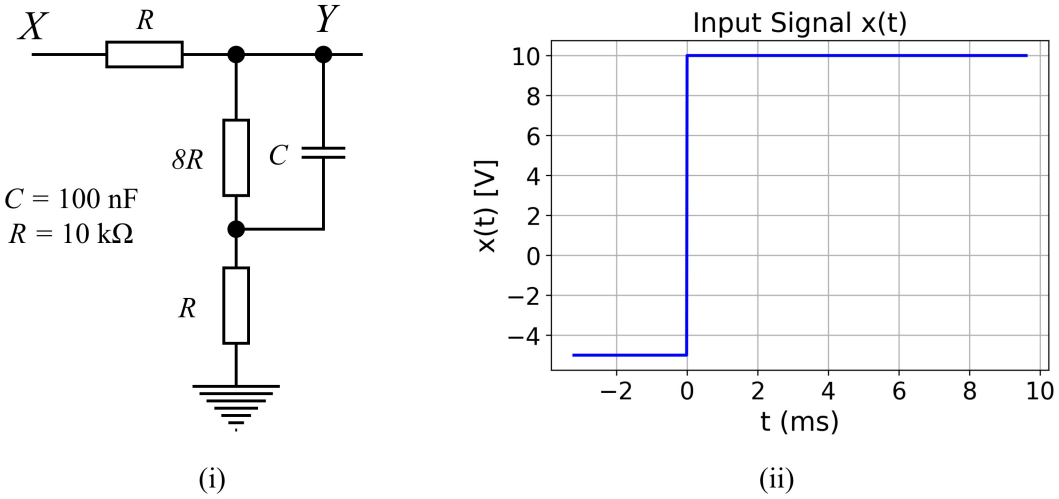


Figure 1c

(d) The amplifier in Figure 1d was originally designed to operate from a 12 V supply at room temperature with a gain of -240. Calculate the supply voltage required for it to produce a gain of -120, and the quiescent collector voltage under that new supply voltage. [7]

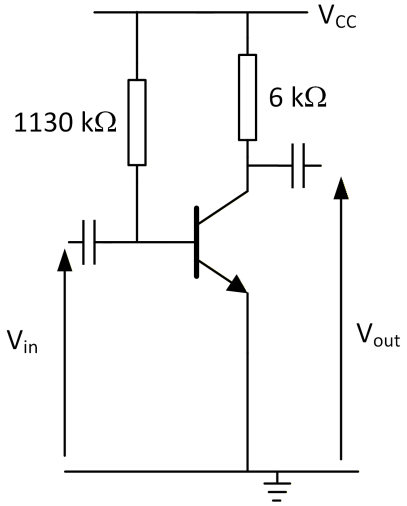


Figure 1d

(e) Calculate the small signal gain of the circuit shown in Figure 1e. Assume R_G is large. [6]

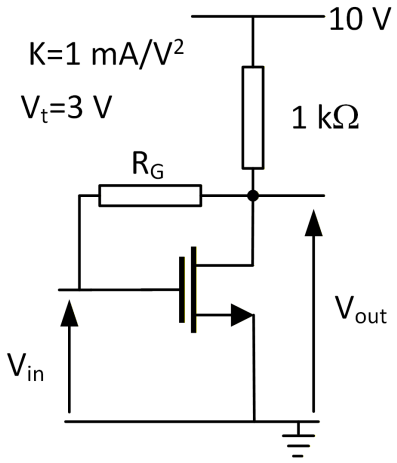


Figure 1e

(f) Explain why the biasing scheme shown in the circuit on the left of Figure 1f is a poor practical solution, but the scheme shown in the circuit on the right (assuming the appropriate choice of component values) is a good solution. [7]

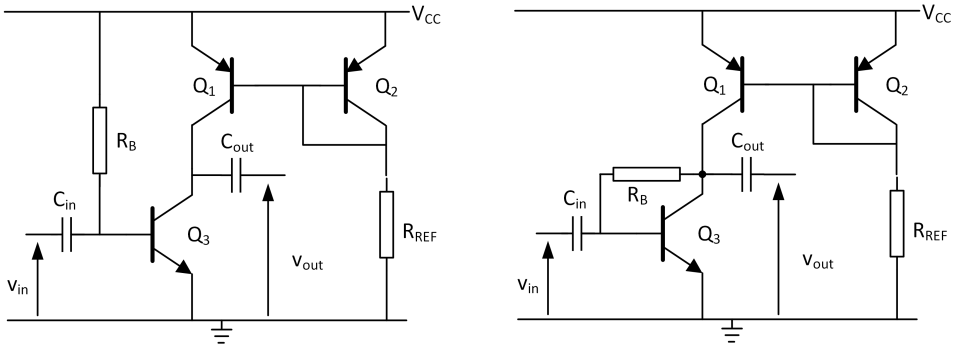


Figure 1f

2. The circuit shown in Figure 2 is an active filter designed for speech processing. Using this circuit, answer the following questions.

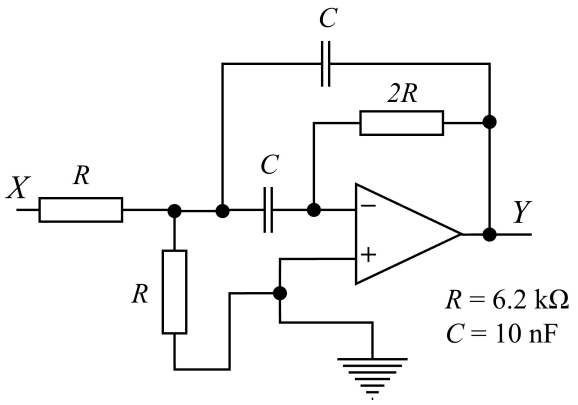


Figure 2

(a) Assuming that the operational amplifier is ideal, use the golden rules of the op-amp to show that the transfer function of the circuit is [8]

$$H(j\omega) = \frac{Y}{X} = -\frac{j\omega RC}{(j\omega RC)^2 + j\omega RC + 1}.$$

(b) From the transfer function $H(j\omega)$ given in Part (a), find expressions for the low-frequency and high-frequency asymptotes. Also, determine the corner frequency in both radians per second (rad/s) and hertz (Hz). [6]

(c) By comparing the denominator of $H(j\omega)$ with the standard second-order form, [4]

$$\left(\frac{j\omega}{\omega_0}\right)^2 + 2\zeta\left(\frac{j\omega}{\omega_0}\right) + 1,$$

determine the damping ratio ζ and hence the quality factor Q, given that $Q = \frac{1}{2\zeta}$.

(d) Based on your answers to Parts (b) and (c), state whether the filter is critically damped, underdamped, or overdamped. Also, comment on whether the corner frequency is suitable for use in a speech-processing filter. [4]

(e) Using the results from Part (b) and (c), sketch a straight-line (asymptotic) approximation of the frequency response between 100 Hz and 100 kHz. Only the magnitude plot is required. [8]

3. A differential amplifier shown in Figure 3. Assume all MOSFETs are identical.

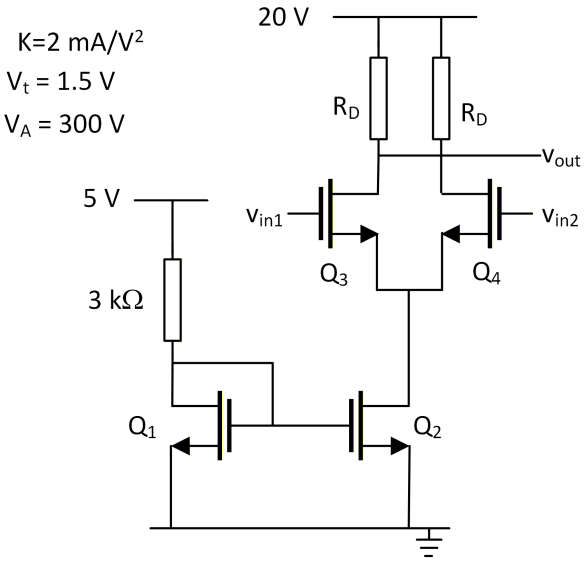


Figure 3

(a) Explain how the differential amplifier operates. In your answer be sure to describe how the circuit can amplify differential signals and (largely) reject common mode signals. [5]

(b) Calculate the following (you may ignore, i.e. assume infinite, the output resistance of the individual transistors when calculating quiescent conditions): [3+2+2]

i. The tail current and the drain current in transistors Q_3 and Q_4 under quiescent conditions.

ii. Calculate the value of R_D to set the output voltage at 12 V when $V_{in1} = V_{in2}$.

iii. Calculate the total quiescent power consumption.

(c) Calculate the differential gain of the amplifier to the single ended output V_{out} when the differential voltage is defined as $V_{in1} - V_{in2}$. [3]

(d) It is suggested that Q_3 and Q_4 are replaced with BJTs, keeping all other components the same. What will the differential gain be? The early voltage of the proposed BJTs is the same as the MOSFETs. Comment on your answer. [5]

(e) It is proposed to reduce the output impedance of a source ($R_{out1} = R_S$), shown on the left of Figure 3e, by adding an emitter follower, shown on the right of Figure 3e. Draw a small signal model of the circuit on the right and derive an expression for its output impedance, R_{out2}. You may assume the BJT is in the forward active mode. [5]

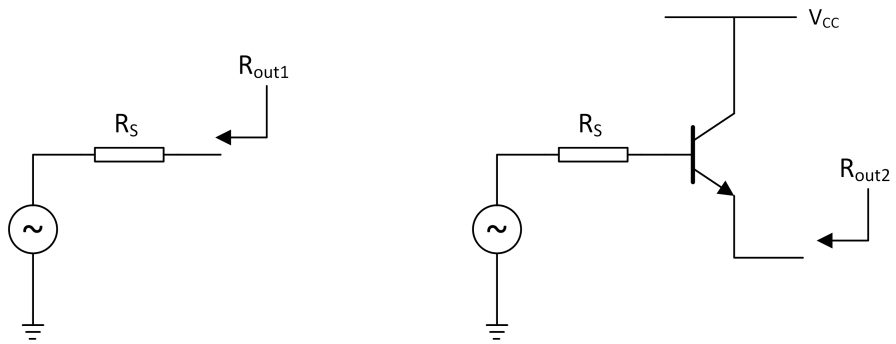


Figure 3e

(f) The emitter follower can only drive positive currents into the load. Suggest an alternative output stage that is able to drive both positive and negative currents into the load and explain its operation. [5]